Submission Worksheet

Submission Data

Course: IT202-008-S2026
Assignment: IT202 PHP Multi-Dimension Problems
Student: Kurt C. (kc784)
Status: Submitted | **Worksheet Progress:** 100%
Potential Grade: 10.00/10.00 (100.00%)
Received Grade: 0.00/10.00 (0.00%)
Started: 3/29/2026 10:31:54 AM
Updated: 3/29/2026 6:17:46 PM
Grading Link: https://learn.ethereallab.app/assignment/v3/IT202-008-S2026/it202-php-multi-dimension-problems/grading/kc784
View Link: https://learn.ethereallab.app/assignment/v3/IT202-008-S2026/it202-php-multi-dimension-problems/view/kc784

Instructions

- Overview Link: https://youtu.be/jd5qiyXReaI

1. Ensure you read all instructions and objectives before starting.
2. Create a new branch from qa called M6-Homework
    1. `git checkout qa` (ensure proper starting branch)
    2. `git pull origin qa` (ensure history is up to date)
    3. `git checkout -b M6-Homework` (create and switch to branch)
3. Copy the template code from here: GitHub Repository - M6 Homework
    - It includes Problems 1-3 and `base.php`. Put all into an M6 folder or similar inside your `public_html`
    - Immediately record to history
        - [ ] `git add public_html`
        - [ ] `git commit -m "adding M6 HW baseline files"`
        - [ ] `git push origin M6-Homework`
        - [ ] Create a Pull Request from M6-Homework to dev and keep it open
4. Fill out the below worksheet
    - Each Problem requires the following as you work
        - [ ] Ensure there's a comment with your UCID, date, and brief summary of how the problem was solved
        - [ ] Initial outline/plan of how you'll solve it via comments (add/commit after this stage)
        - [ ] Code solution (add/commit periodically as needed)
5. Once finished, click "Submit and Export"
6. Locally add the generated PDF to a folder of your choosing inside your repository folder and move it to Github
    1. `git add .`
    2. `git commit -m "adding PDF"`
    3. `git push origin M6-Homework`
    4. On Github merge the pull request from M6-Homework to qa
    5. On Github create a pull request from qa to `prod` and immediately merge. (This will trigger the `prod` deploy to make the render.com prod links work)

7. Upload the same PDF to Canvas
8. Sync Local
    1. `git checkout qa`
    2. `git pull origin qa`

Section #1: (2.5 pts.) Problem 1 - Subset

Progress: 100%

≔ Task #1 (2.50 pts.) - Edit the `processBirds` function to extract properties

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge: Extract the name, color, region of each bird into the $subset array
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

🖼 Part 1:

Progress: 100%

Details:
Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)

```php
// Start edits

// kc784 | 3/29
// plan
// 1. loop through each bird in each array
// 2. pull out only the name, color, region
// 3. add result array to $subset

foreach ($birds as $bird) { // iterates for each bird
    $subset[] = [ // appends entries to $subset
        "name" => $bird["name"], // takes name of bird
        "color" => $bird["color"], // takes color of bird
        "region" => $bird["region"] // takes region from bird
    ]; <-- #39-43 $subset[] =
} <-- #38-44 foreach ($birds as $bird)

// End edits
echo "<pre>" . var_export($subset, true) . "</pre>";
```

code solution



output of program

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Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in `.php`, , zero credit if it does not)
- Direct link to the file on Render.com Prod (Just grab the base prod url and manually enter the path to the file)

URL #1
https://github.com/kurtjcastro/kc784-

it202-[illegible]public_html/M6/problem1.php

URL
https://github.com/kurtjcastro/kc

URL #2
https://kc784-it202-008-prod.onrender.com/M6/problem1.php

URL
https://kc784-it202-008-prod.onre

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Part 3:

Progress: 100%

Details:
Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

With the foreach loop, each bird is looped through from the $birds array, where each iteration, a resulting array is created with only the name, color, and region keys, and appended to $subset. At the end of the loop, only the $subset holds the data from the three fields for each bird.

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Section #2: (2.5 pts.) Problem 2 - Adding Properties

Progress: 100%

≡ Task #1 (2.50 pts.) - Edit the `processCars` function to add properties

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge 1: Add a new property called age that's set from today's year and the car's year
- Challenge 2: Add a new property called `isClassic` that's true/false based on `$classic_age`
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

Part 1:

Progress: 100%

Details:
Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)

```php
// Start edits
// kc704 | 3/29
// plan
// 1. get current year w/ date()
// 2. iterate through each car
// 3. calc age by subtracting car year from current year
// 4. determine isClassic if age is >= classic age
// 5. add original properties w/ age and isClassic to $processedCars

$currentYear    (int)date("Y"); // currat year is taken as int

foreach ($cars as $car) { // iterates over cars
    $age = $currentYear - $car["year"]; // calculates age of car
    $isClassic = $age >= $classic_age; // only true if age of car >= classic car age

    $processedCars[] = array_merge($car, [ // takes all original car properties
        "age" => $age, // adds calculated age
        "isClassic" => $isClassic // adds classic bool
    ]);
} // end foreach ($cars as $car)
// End edits
echo "<pre>" . var_export($processedCars, true) . "</pre>";
```

code solution



output of program

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Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in `.php`, , zero credit if it does not)
- Direct link to the file on Render.com Prod (Just grab the base prod url and manually enter the path to the file)

URL #1
https://github.com/kurtjcastro/kc784-

it202-[redacted]public_html/M6/problem2.php

URL
https://github.com/kurtjcastro/kc

URL #2
https://kc784-it202-008-prod.onrender.com/M6/problem2.php

URL
https://kc784-it202-008-prod.onre

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Part 3:

Progress: 100%

Details:
Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

The current year is taken with the date("Y") and is cast as an int then using a foreach loop, it iterates over each car in the arrays and calculates age by subtracting the car year from the current year. isClassic is only true if the car age is equal to or exceeds the classic car age of 25. The array_merge() then combines the original car properties along with the 2 new results and adds it to $processedCars.

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Section #3: (2.5 pts.) Problem 3 - Join

Progress: 100%

Task #1 (2.50 pts.) - Edit the `joinArrays` function to combine two arrays based on a common key

Progress: 100%

Details:

- Only make edits where noted via provided comments
- Challenge: Combine the data in both arrays by the `userId` property
- Step 1: sketch out plan using comments (include ucid and date)
- Step 2: Add/commit your outline of comments (required for full credit)
- Step 3: Add code to solve the problem (add/commit as needed)

Part 1:

Progress: 100%

Details:
Two screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment)
2. Full output of executing the program (visit the proper file on Render.com qa after a manual deploy)

```php
    // kc784 3/29
    // plan
    // 1. loop through each user
    // 2. loop through $activities to find matching userId
    // 3. merge user + activity into 1 array if id matches
    // 4. put merged result to $joined

    foreach ($users as $user) { // iterates for each user
        foreach ($activities as $activity) { // looks for match in each activity
            if ($user["userId"] == $activity["userId"]) { // joins if userId matches
                $joined[] = array_merge($user, $activity); // combines user + activity as one entry
            }
        } <-- #62-66 foreach ($activities as $activity)
    } <-- #61-67 foreach ($users as $user)
    // end edits
    echo "<pre>" . var_export($joined, true) . "</pre>";
} <-- #45-70 function joinArrays($users, $activities)
```

code solution



output of program

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Part 2:

Progress: 100%

Details:

- Direct link to the file in the homework related branch from Github (should end in `.php`, , zero credit if it does not)
- Direct link to the file on Render.com Prod (Just grab the base prod url and manually enter

the path to the file)

URL #1
https://github.com/kurtjcastro/kc784-it202-008-2026/public_html/M6/problem3.php

URL
https://github.com/kurtjcastro/kc

URL #2
https://kc784-it202-008-prod.onrender.com/M6/problem3.php

URL
https://kc784-it202-008-prod.onre

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Part 3:

Progress: 100%

Details:
Briefly explain how your solution works. Describe the logic, steps, or decisions your code makes to reach the result. Do not restate the problem or list requirements.

Your Response:

For this problem, I used 2 nested foreach loops where the first loop iterates through each user, and the inner loop iterates over each activity. The userId from both arrays is then compared and when a match is found, array_merge take both the user and activity to a single array and appends the results to $joined.

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Section #4: (2.5 pts.) Misc

Progress: 100%

Task #1 (0.83 pts.) - Github Details

Progress: 100%

Part 1:

Progress: 100%

Details:
From the Commits tab of the Pull Request screenshot the commit history Following minimum should be present: addition of baseline files, comments planning thoughts, solutions





commit history

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Part 2:

Progress: 100%

Details:
Include the link to the Pull Request (should end in `/pull/#`, zero credit if it does not)

URL #1
https://github.com/kurtjcastro/kc784-it202-008-2026/pull/79

URL
https://github.com/kurtjcastro/kc

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Task #2 (0.83 pts.) - WakaTime - Activity

Progress: 100%

Details:

- Visit the WakaTime.com Dashboard
- Click `Projects` and find your repository
- Capture the overall time at the top that includes the repository name
- Capture the individual time at the bottom that includes the file time
- Note: The duration isn't relevant for the grade and the visual graphs aren't necessary



overall time





file time

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Task #3 (0.83 pts.) - Reflection

Progress: 100%

Task #1 (0.33 pts.) - What did you learn?

Progress: 100%

Details:
Briefly answer the question with at least a few reasonable sentences

Your Response:

I learned how to extract and organize data from PHP arrays using loops and nested loops in conjunction with functions like array_merge().

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Task #2 (0.33 pts.) - What was the easiest part of the assignment?

Progress: 100%

Details:
Briefly answer the question with at least a few reasonable sentences

Your Response:

The easiest part of the assignment was problem 1 because it was a basic extraction problem that I figured out quickly, since it asked me to extract specific keys from the array into a result holder.

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Task #3 (0.33 pts.) - What was the hardest part of the

assignment?

Progress: 100%

Details:
Briefly answer the question with at least a few reasonable sentences

Your Response:

The challenging part of this assignment was problem 3 because of the use of a nested loop to find matching IDs from both the users and the activities. The results are then combined once a match is found.

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